FloodLens Nepal: satellite-based flood and debris mapping for the August 2026 Trishuli event

Track B, Multimodal AI Hackathon 2026 (IIT Mandi) · [[FILL: team name and members]] · Educational prototype, not operational emergency guidance.

How to use this draft: every number below comes from run 4849e43e (`summary.json`) or `metrics.json`.

1. Summary

FloodLens takes an area and a flood date and answers three questions for rescue teams using only free satellite and map data: where did the flood and debris hit, which pre-event OpenStreetMap (OSM) buildings, roads and bridges are potentially affected, and which settlements are potentially cut off from the nearest hospital or town. It runs end to end from raw Sentinel-1 data through a command line and a Streamlit dashboard. Every number is written to one file, `summary.json`, and the situation report is generated from that file only.

Case study (run 4849e43e). For a 189 km² Langtang-area AOI in Rasuwa district (Bhote Koshi–Trishuli drainage) and the 26 August 2026 event, the satellite-derived estimate is **0.58 km² of likely flood/debris change** (0.36 km² water, 0.22 km² debris) at medium confidence, from radar only. OSM roads and health facilities could not be downloaded for this run, so cut-off settlements were **not assessed**. A quantitative comparison with the Copernicus EMSR927 reference maps was not possible, because the reference data we obtained does not overlap the analysed area (section 6).

2. Data and compliance

Input	What we used
Sentinel-1 GRD (IW, VV+VH)	Before: 16 Aug 2026 12:21 UTC; after: 28 Aug 2026 12:21 UTC; relative orbit 85, ascending (same track, 12 days apart)
Sentinel-2 L2A	Post-event scene of 27 Aug 2026, 78.5% scene-level cloud: no usable optical confirmation

Elevation	Copernicus DEM GLO-30
OSM	Pre-event snapshot dated 27 Jul 2026, the latest date available. Retrieved through the Overpass historical (attic) date feature because the ohsome geometry endpoint returned HTTP 403 from our networks
Training data	Kuro Siwo (Bountos et al., 2024)

Rules and how they are enforced. Reference maps (EMSR927) and any post-event data live in a separate `validation_only/` folder that pipeline code never imports. Automated tests check that no pipeline file references validation data, that every OSM request carries a date strictly before the event, that the two Sentinel-1 scenes share relative orbit and direction, and that every number in the report exists in `summary.json`. The configuration was frozen before EMSR927 was opened (config hash `a922dc75…`, recorded in the run), so no threshold was tuned on the reference event.

3. Method

3.1 Scene selection. The selector searches the Copernicus Data Space catalogue and accepts only a same-orbit, same-direction, same-mode Sentinel-1 pair that straddles the event date. If none exists it stops, states the exact reason and lists candidate scenes. It never mixes tracks.

3.2 Sentinel-1 processing. Terrain-corrected sigma-0 backscatter is requested server-side from the Copernicus Data Space Process API with the Copernicus DEM, on one common 10 m UTM grid for all scenes. Speckle is reduced in linear power; decibels are used only for change maths. Co-registration check: [[FILL: measured shift in metres. The `sar_alignment_rmse_m` field in summary.json is a 0.0 placeholder and must not be quoted]].

3.3 Terrain masks. From the DEM we derive slope, radar layover and shadow, and height above nearest drainage (HAND). Pixels on steep slopes, in layover or shadow, or far above the channel are excluded from flood candidates. [[CHECK detection_funnel.json: confirm HAND was real and not the local-relief proxy in this run]].

3.4 Baseline: two-sided change detection. Water is a strong VV decrease with a dark post-event VV; debris is a VH and VV increase (a rough deposit). Thresholds adapt to the measured noise level of the scene, strong seed pixels are grown by hysteresis, permanent water is excluded, and small patches are removed. Every filtering stage is counted in `detection_funnel.json`.

3.5 AI component: U-Net. A U-Net with a ResNet-18 encoder is trained on Kuro Siwo flood-water labels. Input is six channels: pre-event VV and VH, post-event VV and VH, and the two post-minus-pre differences, all in dB and normalised with training statistics. Training used a binary cross-entropy plus Dice loss masked to valid pixels, up to 4,000 patches, 15 epochs on a Colab T4 GPU, with events split between training, validation and test (never random patches). It is trained from scratch: ImageNet weights were deliberately excluded because that pretraining data is not in the permitted list. Inference runs at the native 10 m grid used in training. Because Kuro Siwo labels are flood water, **water comes from the U-Net and debris from the baseline rule**.

3.6 Optical confirmation. On clear Sentinel-2 pixels, MNDWI confirms water and NDVI drop confirms debris, and each candidate is labelled SAR-only, confirmed or contradicted. For this event there was too little clear coverage, so every candidate is SAR-only. No cloud-removal or reconstruction model is used anywhere.

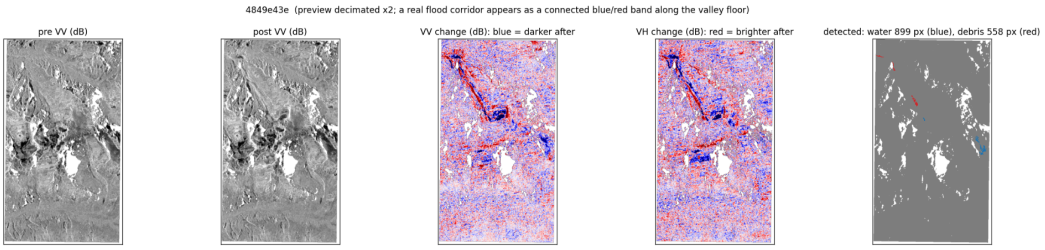
3.7 Impact and cut-off analysis. Pre-event buildings are flagged when at least 10% of their footprint lies in the mask; roads and bridges when they intersect it (roads need a minimum intersection length to ignore slivers). For access, a motorable-road graph before the event (G0) is compared with the same graph with hazard-intersecting sections removed (G1) at several hazard buffers. Settlements and hospitals are snapped to the nearest road node within a maximum distance. A settlement is *connected*, *potentially cut off* (reachable in G0, not in G1), or *uncertain* (poor snapping, disconnected at baseline, or a class that changes with the buffer). A road graph that is already disconnected before the event is never called flood isolation.

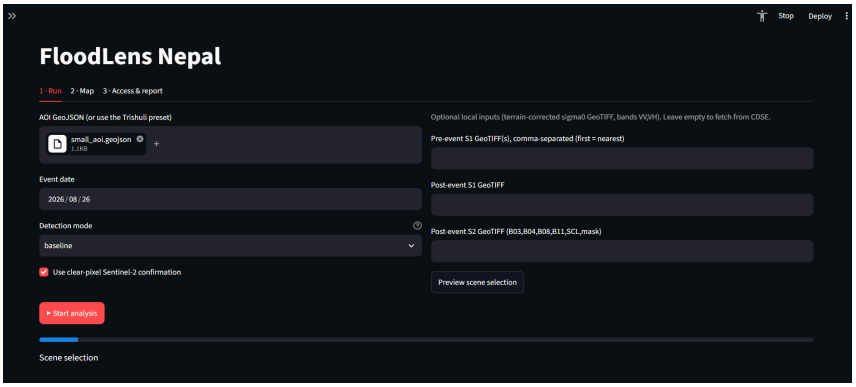
3.8 Report. A template fills the situation report from `summary.json` only, with approved wording: "satellite-derived estimate", "likely flood/debris change area", "potentially affected", "potentially cut off based on mapped-road connectivity", "requires field verification".

4. Results for the case study (run 4849e43e)

Item	Value
AOI	189.0 km², Langtang area, event date 26 Aug 2026
Sentinel-1 pair	16 Aug → 28 Aug 2026, orbit 85 ascending, same track: yes
Likely flood/debris change area	0.58 km² (water 0.36 km² = 3,593 px; debris 0.22 km² = 2,176 px at 10 m)

Confidence	Medium
Optical confirmation	None (cloud)
OSM layers retrieved	buildings 519, bridges 10, settlements 16, towns 2
OSM layers not retrieved	roads (23 of 33 tiles failed) and health facilities (33 of 33): public server returned HTTP 504
Potentially affected	0 of the 519 retrieved buildings and 0 of the 10 retrieved bridges overlap the mask; road length not assessed
Potentially cut-off settlements	Not assessed (no roads or hospitals available)
Run time	2,723 s.





Zero affected buildings and bridges means none of the *retrieved* OSM features overlapped the mask. It is not evidence of no damage, and a mask cannot show structural damage.

5. Evaluation of the AI component

Evaluation set	IoU	F1	Precision	Recall	Area difference
Validation events (used to choose the best epoch)	0.729	0.844	0.905	0.790	−12.8%
Held-out test events (never used in training or selection)	0.617	0.763	0.670	0.887	+32.4%
Himalayan unseen event, EMSR927 (check-only)	Not possible: no overlap with the reference (section 6)				

On unseen events the model finds most flood water (recall 0.89) but over-predicts it precision 0.67, +32% area). Validation IoU moved between 0.64 and 0.73 from epoch to epoch because there are few validation events, so the best-epoch choice is noisy and the test row is the honest figure. The held-out Kuro Siwo events are mostly outside the Himalaya, so these numbers are **not** a measure of Himalayan performance; an EMSR927 comparison would have addressed this, but it could not be completed (section 6). Inference took about 0.011–0.013 s per 224×224 patch on the T4 GPU. We did not build an ensemble because it was not validated on held-out data.

In a development run on this area the radar change images showed a connected band of strong darkening along the valley, but the model responded mainly to broad water and only weakly along the narrow channel, which is only a few pixels wide

6. Comparison with Copernicus EMSR927 (check-only)

No quantitative comparison with EMSR927 was possible for the Langtang run. The configuration was frozen before any EMSR927 file was opened, and the check-only script accepted the frozen configuration. It then found no pixels to evaluate, because the reference data we obtained lies outside the area we analysed.

- Our run (4849e43e) covers lon 85.465–85.58 and lat 28.185–28.349 (Langtang area).
- EMSR927 AOI02 (Timure): its only mapped event is one photo-interpreted landslide of 1.29 km² at lon 85.357–85.38 and lat 28.242–28.28, about 8 km west of our area. It is a landslide, not a flood extent.
- A second file, EMSR927-aois.geojson (area-of-interest outlines, no event polygons), also gave no overlap. Script result: 5,769 predicted flood pixels, 0 reference flood pixels, 0 pixels inside the reference extent, 0 evaluated.

Consequence. Performance on Himalayan scenes is therefore not validated against EMSR927. The only quantitative evidence is the held-out Kuro Siwo test (IoU 0.617, F1 0.763), which is mostly non-Himalayan. This is a known gap in the case study. It could be closed by running the same frozen configuration on an area that overlaps an EMSR927 event layer (for example the Bhote Koshi corridor near Timure and Syabrubesi) and scoring it with the unchanged script; we did not complete this in the time available. Even then, the AOI02 reference is a single small landslide, so it would check the debris class only.

7. Limitations: what the system cannot do

- **No early warning.** Sentinel satellites revisit every few days, so the system cannot warn minutes ahead of a sudden glacier collapse; it maps impact afterwards.
- **Radar artefacts.** C-band radar gives false positives on steep terrain, in shadow and layover and on wet soil, and cannot see water under dense vegetation.
- **Narrow channels and debris.** Narrow flood channels are under-detected at 10 m. The U-Net was trained on broad flood water mostly outside the Himalaya, and debris is mapped by the radar baseline only, with no debris labels to validate it.
- **Optical data is cloud-limited.** For this event it gave no confirmation. No reconstructed pixels are used.
- **A mask is not damage.** A flood/debris mask cannot confirm structural damage to any building or bridge.
- **OSM data.** For this run roads and hospitals could not be retrieved (public server errors), so no cut-off result exists. OSM may also omit roads, bridges, buildings, hospitals or settlements, the snapshot is dated 27 Jul 2026 (the latest available, a month before the event), and coverage of the Overpass historical database was not independently verified. A graph path does not mean a hospital is operational or a road passable.
- **Speed and dependence on public services.** The case-study run took 2,723 s, well above a live-demonstration slot, and relies on public servers (the ohsome geometry endpoint was blocked with HTTP 403 and the Overpass server was overloaded). A queue-aware, tiled and cached client reduces but does not remove this.

- **Evidence base.** No quantitative comparison with an EMSR927 reference was possible, so performance on Himalayan scenes is not validated; thresholds are not validated on other Himalayan events.
- **Use.** Estimates support prioritisation, are not an official assessment, and require field verification.

8. Development Workflow & External AI Tooling

• To accelerate pipeline development, harden geospatial query modules, and synthesize disaster response documentation within the 15-day sprint window, LaLaLand integrated state-of-the-art AI assistant platforms:

1. Anthropic (Claude 3.5 Sonnet / Claude Pro): Utilized as the primary architectural reasoning engine for writing core Python modules (e.g., Overpass API caching layers, Rasterio spatial clipping routines, and NetworkX shortest-path graph traversals), as well as debugging U-Net tensor shapes and memory bottlenecks.
2. Perplexity AI: Deployed for rapid technical research on synthetic aperture radar (SAR) backscatter calibration formulas, literature review of the KuroSiwo dataset, and verifying geographic coordinates and historical disaster reports for the Trishuli river basin

9. Responsible use

This is an educational prototype. No victim imagery is used. Wording avoids "destroyed" or "guaranteed blocked", and a persistent banner states that results are not operational guidance.

10. Reproducibility

Code and README: https://github.com/smansipillai009/FloodLens. Setup, `pytest`, the CLI (`run.py search`, `osm`, `analyse`), the dashboard and the training and validation commands are in the README. The trained checkpoint, normalisation statistics and `metrics.json` are included. Each run writes a manifest with scene IDs, OSM snapshot date, configuration hash, versions and timestamps.

11. Attributions and references

Contains modified Copernicus Sentinel data 2026.

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Training data: Bountos, N. I. et al. (2024). Kuro Siwo: 33 billion m² under the water! A global multi-temporal satellite dataset for rapid flood mapping. NeurIPS Datasets and Benchmarks.

European Union, Copernicus Emergency Management Service data (EMSR927, used for comparison only, never as an input).